



TENDER NOTICE FOR MNRE II PROJECT ON 40 KWTH SOLAR THERMAL

PART I

SOLAR THERMAL STEAM GENERATION SYSTEM

RKDF University is undertaking a Solar Thermal Storage Project in Collaboration/ co-operation with a US based Institute, RPI New York.

Parties interested in participating are requested to submit a quote for 40 kWth Peak Solar Concentrating System with stationary focus. The concentrators should have stationary focus and should generate above 500 degree Celsius temperature in focus of its concentrators at the core of receiver.

The supplier will have to interact with Co-PI of the project based in USA (Dr. Partha S Dutta) to provide him info needed for storage material that will be provided through RKDF, University, Bhopal India. RPI will design a Solar Thermal Storage System to be able to generate steam at 200 degree C + temperature, sufficient to regenerate solvent in the adjacent CCS Test Unit or generate steam equivalent to run 10 TR VAM machine/ Turbine

(Both CCS plant and VAM machine/ Steam Turbine are out of scope of MNRE II Project).

The project has to be designed, supplied and installed and completed in 6 months including installation and handing over the system after training the team of RKDF.

PART II

SOLAR THERMAL ENERGY STORAGE

Tender is solicited for a Solar Thermal Energy Storage (TES) system to capture heat from steam flowing through Scheffler Dish receivers in the normal temperature range of 200-250 degrees C and peak temperature of over 500 degrees C. Tender is requested for the following 3 options for the TOTAL capacity for the TES system: 5.5 KWth (TES 1), 40 KWth (TES 2), 320 KWth (TES 3). All TES systems must be static.

Tender is requested for the following 3 options

- a) TES system: 5.0 KWth (TES 1),
- b) 40 KWth (TES 2),
- c) 320 KWth (TES 3).

All TES systems must be must have stationary focus.

Option 2 a)

The TES1 must be coupled with a singular Scheffler Dish of 16 sq. m collection area collecting solar energy for 1 hour during the day. The TES 2 must be coupled with 8 Scheffler Dishes (16 sq.m. area each) connected in series collecting solar energy for 1 hour during the day. The TES 3 must be coupled with 8 Scheffler Dishes as in TES 2 collecting solar energy for 8 hours during the day. Duration of the thermal energy storage must be 8-10 hours after collection phase concludes (if energy extraction is not initiated during this period). TES systems must include heat exchangers and mechanisms for efficient thermal energy capture and delivery.

Option 2 b)

The following are the main TES system components: Solid phase TES media, Storage Vessel, Solid phase thermal insulation media, Solenoids, Control valves, VFD, Blower and Motor, Heat Exchanger, Tubing, Fittings, Framing, Computer, Instrumentation, and Data Acquisition System. The cost proposal must include all the associated expenses for engineering design, fabrication, installation and testing of the TES. The expense categories include: hardware, software, engineering design, labor costs.

The project duration from the initiation of the project is 18 months.

Parties have to quote for both, Part I & II (option 2-a and /or option 2-b) separately (Steam Generation & TES parts).

Parties quoting only for Part I will not be considered, they will disqualify in technical scrutiny of the tender itself.

The parties quoting have to submit the Bids in two envelopes the first envelope should consist of Technical details, qualification and experience of the company and the second envelope should consist of financial bid. The envelopes should be sent by Speed Post or Courier.

IMPORTANT DATES:

- Last date of submission of Tender: 12 June 2019, 1500 hrs.
- Both Technical & Financial Bids shall be opened on 15 June 2019 at 11 AM
- All correspondence by mails only

TIMES OF INDIA ADVERTISEMENT DATED 25 MAY 2019

**RAMKRISHNA DHARMARTH FOUNDATION UNIVERSITY, AIRPORT BYPASS
ROAD, GANDHI NAGAR, Near RGPV, BHOPAL- 462033**



TENDER ENQUIRY- (MNRE/ RKDF/ RPI / 01: Date 29 May 2019)

**Sealed Tenders are invited for Turn-key job of 'SOLAR THERMAL PROJECT (40 KWth)
INTEGRATED WITH THERMAL ENERGY STORAGE' sponsored by MNRE, GOI**

BROAD

PART I: Engineering design, installation and testing of pilot plant at RKDF University , Bhopal with 40kWth (thermal) capacity integrated with 24x7 thermal energy storage (TES) Steam Generation Solar Concentrating System shall be consisting of 8 Scheffler Disc with auto tracking having stationary focus. The concentrators should generate above 500 degree Celsius temperature in focus of its concentrators, complete with steam piping, insulation and safety devices.

Part II: Tender is also solicited for a Solar Thermal Energy Storage (TES) systems to capture heat from steam flowing through Scheffler Dish receivers in the normal temperature range of 200-250 degrees C and peak temperature of over 500 degrees C. Tender is requested for the following 3 options for the TOTAL capacity for the TES system: 5.5 KWth (TES 1), 40 KWth (TES 2), 320 KWth (TES 3). All TES systems must be static. System should be fully equipped with instrumentation for R&D work of field testing of energy storage capacity, steam generation and electric power out-put. (TES TECHNOLOGY IN COLLABORATION WITH RPI, USA)

SPECIFICATIONS:

Both Technical & Financial bids (Part I & II both together) should be submitted in separate sealed envelopes at the above address through speed post / registered post/ courier before 12 June 2019 (3:30 PM). Sealed envelope covers should be marked with "TENDER NO: MNRE/ RKDF/ RPI / 01". Both bids will be opened on 15 June 2019. Offer only for Part I will not be considered.

Note: For any clarification contact on e-mail vc@rkdf.ac.in & for details of the project and tendering of Part-I & II please refer our website rkdf.ac.in

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